



Database Proposal for Drug Testing Laboratory

**Ryan Bennett, Chirag Chirag,
Christina Kuleshevich, Anusuya Roysarkar,
Emily Tellschow**

Introduction

PURPOSE

Manage and track essential information related to clients, employees, tests conducted, and laboratory equipment



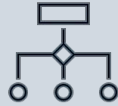
GOAL

Improve operational efficiency of the drug testing laboratory



Benefits of the Database

Streamline data
management



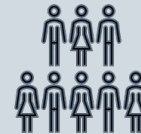
Enhance tracking of
client information



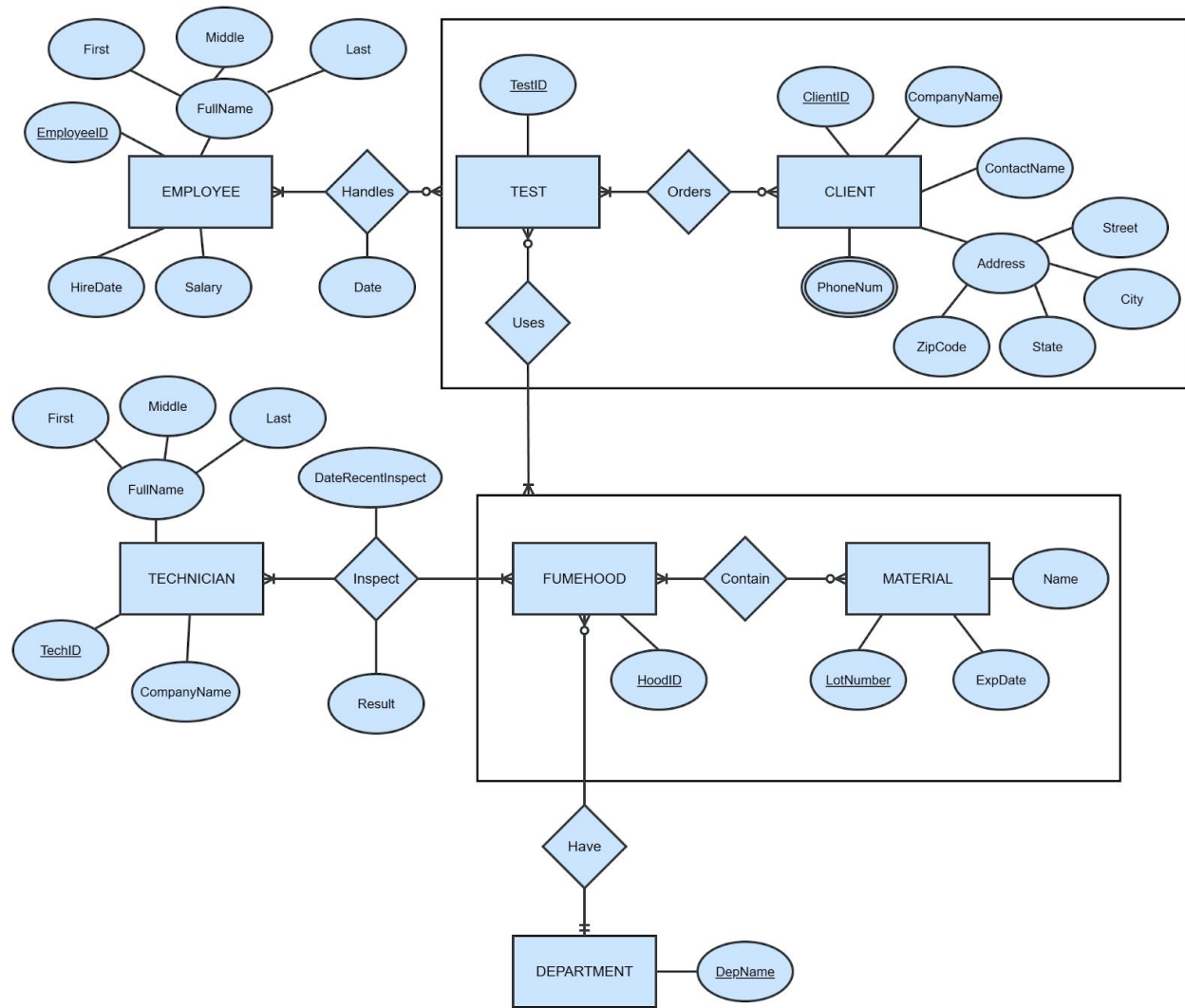
Ensure compliance with
regulatory standards



Improve employee
management



Entity-Relationship Diagram



Relational Schema Mapping

1. EMPLOYEE (EmployeeID, First, Middle, Last, HireDate, Salary)
2. TEST (TestID)
3. CLIENT (ClientID, CompanyName, ContactName, Street, City, State, ZipCode)
 CLIENTPHONENUM (PhoneNum, ClientID[FK])
4. ORDERS (TestID[FK], ClientID[FK])
5. HANDLES (EmployeeID[FK], TestID[FK], ClientID[FK], Date)
6. FUMEHOOD (HoodID, DepName[FK])
7. DEPARTMENT (DepName)
8. MATERIAL (LotNumber, Name, ExpDate)
9. CONTAIN (HoodID[FK], LotNumber[FK])
10. USES (TestID[FK], HoodID[FK], LotNumber[FK])
11. TECHNICIAN (TechID, First, Middle, Last, CompanyName)
12. INSPECT (TechID[FK], HoodID[FK], DateRecentInspect, Result)

	A	B	C	D	E
1	EmployeeID	First	Last	HireDate	Salary
2	101	Nevsa	Pennock	3/5/2022	105000
3	102	Corrianne	Andrich	3/8/2022	125000
4	103	Julee	Leverich	3/6/2022	150000
5	104	Cyrille	Upston	4/17/2022	170000
6	105	Deeanne	Blinco	9/15/2022	145000
7	106	Janessa	Catmull	3/8/2023	115000
8	107	Shari	Clowes	8/4/2023	125000
9	108	Arvie	Tiesman	12/26/2023	150000
10	109	Nobie	Rame	2/7/2024	130000
11	110	Ferris	Weson	5/16/2024	150000

	A	B	C	D	E	F	G
1	ClientID	CompanyName	ContactName	Street	City	State	ZipCode
2	CL1	Thoughtblab	Madlin Jacobowits	9792 Grasskamp Pass	Austin	TX	78789
3	CL2	Oyope	Britte Keller	2 Elgar Pass	Washington	DC	20041
4	CL3	Fivechat	Rosalie Gierhard	75 Texas Street	Denver	CO	80565
5	CL4	Oyoyo	Nelia Spicer	6404 Nancy Street	Austin	TX	78744
6	CL5	Fanoodle	Randi Broxholme	7 Old Shore Road	Colorado Springs	CO	80945
7	CL6	Avavee	Catrina Stonier	25 Cordelia Hill	Golden	CO	80580
8	CL7	Leexo	Jesse Jeynes	879 6th Plaza	El Paso	TX	79923
9	CL8	Dynabox	Derrick Eccleshare	32274 Gerald Terrace	Sioux Falls	SD	57110
10	CL9	Twitternation	Taite Dockrey	714 Blue Bill Park Lane	Boulder	CO	80302
11	CL10	Edgetag	Johnathon Novelli	868 Fremont Way	Washington	DC	20067

	A	B	C	D
1	EmployeeID[FK]	TestID[FK]	ClientID[FK]	Date
2	101	T1	CL1	4/5/2024
3	102	T2	CL2	4/8/2024
4	103	T3	CL3	4/6/2024
5	104	T4	CL4	4/12/2024
6	105	T5	CL5	9/12/2024
7	106	T6	CL6	3/8/2025
8	107	T1	CL7	8/4/2025
9	108	T2	CL8	9/4/2025
10	101	T2	CL9	9/7/2025
11	101	T2	CL10	9/8/2025
12	102	T1	CL9	9/9/2025
13	101	T5	CL10	9/10/2025
14	105	T5	CL3	9/12/2025
15	105	T6	CL9	9/12/2025
16	101	T5	CL9	10/1/2025
17	102	T1	CL10	10/3/2025
18	105	T2	CL6	10/7/2025
19	101	T3	CL6	10/11/2025
20	105	T4	CL10	11/12/2025

	A	B	C
1	LotNumber	Name	ExpDate
2	LOT10	Sodium Hydroxide	11/24/2026
3	LOT11	Acetonitrile	5/23/2030
4	LOT12	Methanol	12/14/2025
5	LOT13	Ethanol	3/2/2027
6	LOT14	Phosphate Buffer	11/22/2026
7	LOT15	Nitric Acid	4/13/2026
8	LOT16	Magnesium	6/26/2028
9	LOT17	Phosphorus	10/21/2027
10	LOT18	Silicon	8/24/2027
11	LOT19	Chlorine	4/17/2030

	A	B	C	
1	TechID[FK]	HoodID[FK]	DateRecentInspect	Result
2	TH1	H1	1/11/2025	Pass
3	TH2	H2	3/7/2025	Pass
4	TH3	H3	5/10/2025	Fail
5	TH1	H4	8/6/2025	Pass
6	TH3	H5	9/10/2025	Pass

	A	B	C
1	TestID[FK]	HoodID[FK]	LotNumber[FK]
2	T1	H1	LOT10
3	T2	H2	LOT11
4	T3	H3	LOT12
5	T4	H4	LOT13
6	T5	H5	LOT14
7	T6	H1	LOT15
8	T1	H2	LOT16
9	T2	H3	LOT17
10	T2	H4	LOT18
11	T2	H5	LOT19
12	T1	H1	LOT19
13	T5	H2	LOT19
14	T5	H3	LOT15
15	T6	H3	LOT16
16	T5	H4	LOT19
17	T1	H5	LOT12
18	T2	H5	LOT18
19	T3	H3	LOT17
20	T4	H1	LOT19

Use Case 1:

Find all fume hoods that have a “Fail” result in the most recent inspection table (included their department), -- and what materials/tests rely on them?

Significance:

- Fume hoods that fail inspection cannot be used until they have been repaired.
- SQL database provides an efficient way to identify those hoods that have failed and all relevant information to that hood.

```
SELECT HoodID, DateRecentInspect, Result, TestID, LotNumber, Name  
AS MaterialName FROM INSPECT NATURAL LEFT JOIN FUMEHOOD  
NATURAL LEFT JOIN USES NATURAL LEFT JOIN MATERIAL WHERE Result  
= 'Fail' ORDER BY HoodID, TestID;
```

	HoodID ▲	DateRecentInspect ▲	Result ▲	TestID ▲	LotNumber ▲	MaterialName ▲
1	H3	2025-05-10	Fail	T2	LOT17	Phosphorus
2	H3	2025-05-10	Fail	T3	LOT17	Phosphorus
3	H3	2025-05-10	Fail	T3	LOT12	Methanol
4	H3	2025-05-10	Fail	T5	LOT15	Nitric Acid
5	H3	2025-05-10	Fail	T6	LOT16	Magnesium

Fume Hood Safety Status

Red= Failed Inspection, Green = Passed

DepName	HoodID		Result
Analysis and Reporting	H5	 Pass	 Fail
Sample Processing	H1	 Pass	 Pass
	H3	 Fail	
Testing	H2	 Pass	
	H4	 Pass	

Result broken down by DepName and HoodID. Color shows details about Result. The marks are labeled by Result.

Use Case 2:

Which clients' orders depend on failed fume hoods?

Significance:

- This analysis ensures that no client orders rely on unsafe or non-functional equipment. It allows the lab to quickly notify affected clients, reroute or reschedule impacted tests, and prioritize repairs for the hoods that disrupt the most business. As a result, the lab minimizes delays, maintains service quality, and protects client trust.

SELECT HoodID, TestID, ClientId, CompanyName FROM INSPECT NATURAL LEFT JOIN USES NATURAL LEFT JOIN ORDERS NATURAL LEFT JOIN CLIENT WHERE Result = 'Fail' GROUP BY CompanyName, ClientID, TestID;

	HoodID ▲	TestID ▲	ClientId ▲	CompanyName ▲
1	H3	T2	CL6	Avavee
2	H3	T3	CL6	Avavee
3	H3	T6	CL6	Avavee
4	H3	T2	CL8	Dynabox
5	H3	T2	CL10	Edgetag
6	H3	T5	CL10	Edgetag
7	H3	T5	CL5	Fanoodle
8	H3	T3	CL3	Fivechat
9	H3	T5	CL3	Fivechat
10	H3	T2	CL2	Oyope
11	H3	T3	CL4	Oyoyo
12	H3	T2	CL9	Twitternation
13	H3	T5	CL9	Twitternation
14	H3	T6	CL9	Twitternation

Heatmap – Client × Hood

COMPANY NAME	CLIENT ID	FAILED HOOD	
		Hood H3	
Avavee	CL6		3
Twitternation	CL9		3
Edgetag	CL10		2
Fivechat	CL3		2
Dynabox	CL8		1
Fanoodle	CL5		1
Oyope	CL2		1
Oyoyo	CL4		1

Use Case 3:

Which employees handled the most tests (show EmployeeID, First, Last names, Number of handled tests)?

Significance:

- Significance: Shows which employees are handling the highest testing workload
- Helps managers rebalance assignments and prevent burnout
- Highlights experienced staff who can mentor or train others
- Informs staffing and hiring decisions during peak testing periods

```
SELECT EmployeeID, First, Last, COUNT(TestID) AS  
Numb_Handled_Tests FROM EMPLOYEE NATURAL LEFT JOIN HANDLES  
GROUP BY EmployeeID, First, Last ORDER BY Numb_Handled_Tests  
DESC;
```

	EmployeeID ▲	First ▲	Last ▲	Numb_Handled_Tests ▲
1	101	Nevsa	Pennock	6
2	105	Deeanne	Blinco	5
3	102	Corrianne	Andrich	3
4	108	Arvie	Tiesman	1
5	103	Julee	Leverich	1
6	104	Cyrille	Upston	1
7	106	Janessa	Catmull	1
8	107	Shari	Clowes	1
9	109	Nobie	Rame	0
10	110	Ferris	Weson	0



Use Case 4:

Which clients order the most tests?

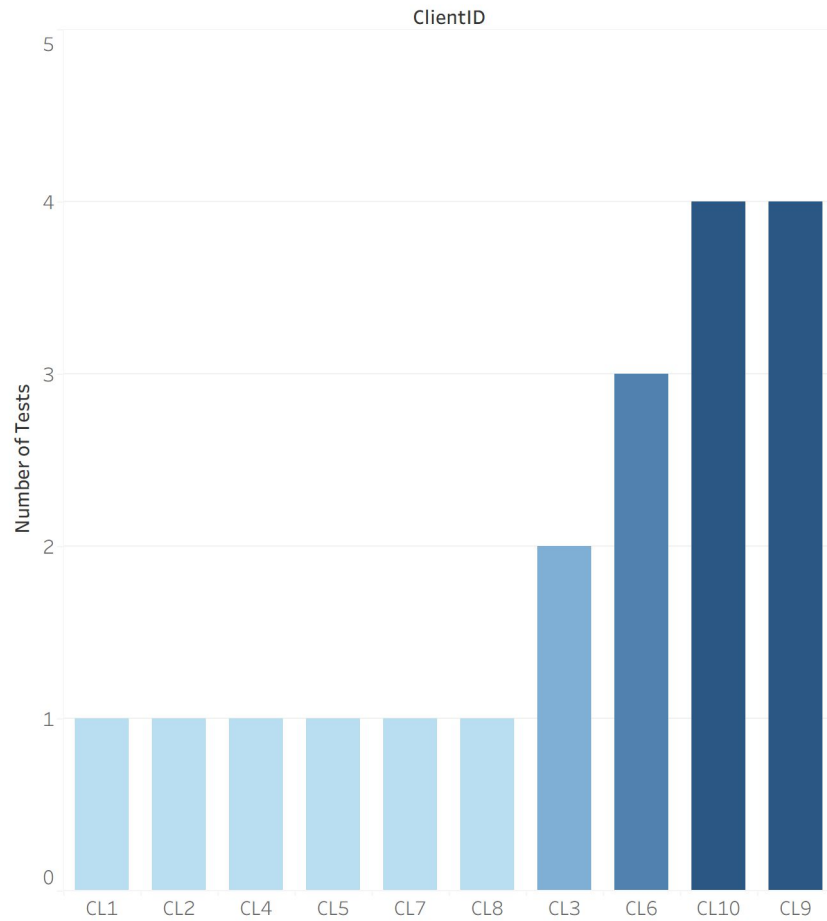
Significance:

- By ranking clients based on the number of tests they order, the lab can see which companies contribute the most to revenue and workload.
- Understanding client activity helps the lab adjust pricing, identify opportunities for upselling, or develop long-term partnerships with the most engaged clients.

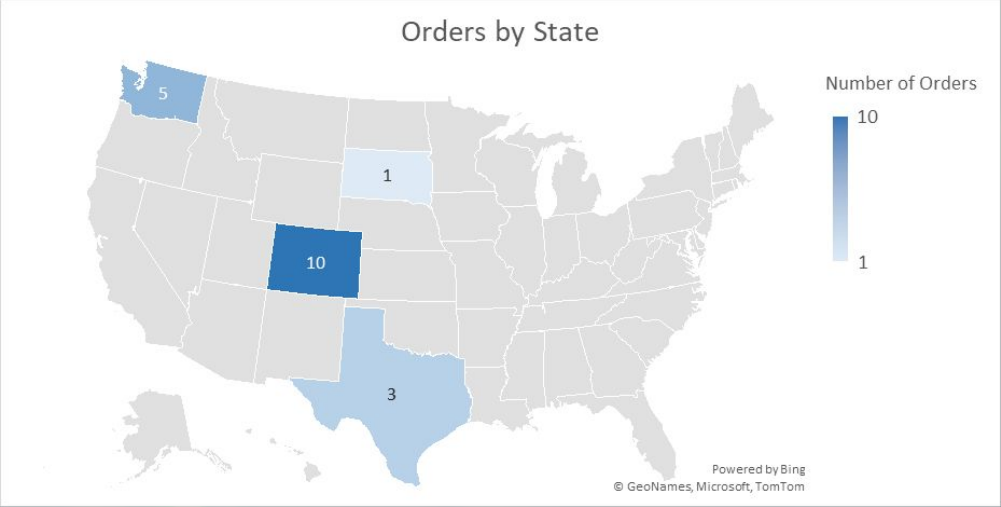
```
SELECT ClientID, CompanyName, COUNT(TestID) AS  
Numb_Ordered_Tests FROM CLIENT NATURAL LEFT JOIN HANDLES  
GROUP BY ClientID, CompanyName ORDER BY Numb_Ordered_Tests  
DESC;
```

	ClientID ▲	CompanyName ▲	Numb_Ordered_Tests ▲
1	CL10	Edgetag	4
2	CL9	Twitternation	4
3	CL6	Avavee	3
4	CL3	Fivechat	2
5	CL1	Thoughtblab	1
6	CL2	Oyope	1
7	CL4	Oyoyo	1
8	CL5	Fanoodle	1
9	CL7	Leexo	1
10	CL8	Dynabox	1

Client Test Order Volume



Sum of Numb_Ordered_Tests for each ClientID. Color shows sum of Numb_Ordered_Tests.



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Use Case 5:

Are some tests ordered by multiple clients (the most popular test)?

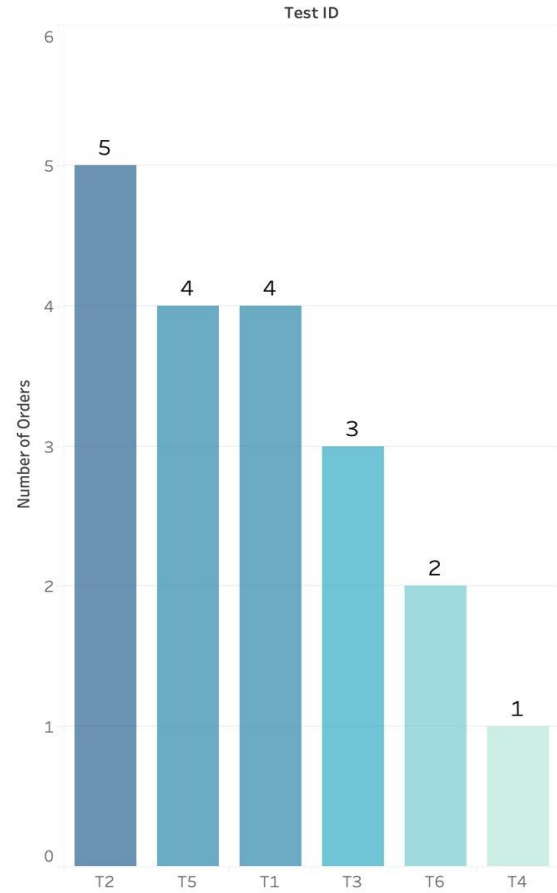
Significance:

- Can be used to influence decisions around employee training
- Useful for ordering materials
- Help determine scheduling and resource allocation

```
SELECT TestID, COUNT(ClientID) AS Numb_Ordered_Tests FROM  
HANDLES GROUP BY TestID ORDER BY Numb_Ordered_Tests DESC;
```

	TestID ▲	Numb_Ordered_Tests ▲
1	T2	5
2	T1	4
3	T5	4
4	T3	2
5	T4	2
6	T6	2

Most Frequently Ordered Tests



Takeaways

Data-driven decision making



Regulatory compliance assurance



Reduced errors in data entry



Increase staff collaboration

